Manufacturing processes, industrialization

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IDENTIFICATION

CODE : GI-3-S1-EC-PFI
ECTS : 3

HOURS

Cours :	0h
TD :	40h
TP :	8h
Projet :	0h
Evaluation :	0h
Face à face pédagogique :	48h
Travail personnel :	0h
Total :	48h

ASSESMENT METHOD

Written Assessment
Supervised Practical Work

TEACHING AIDS

Teaching Method
A project provides a common thread
- 1 lecture session followed by 2 tutorial sessions
- Work in groups of 2 people
The practical work is organized in groups of 3 students and takes place on a machine tool

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Conception de produits et systèmes industriels (GI-3-S1-UE-CPSI) and contributes to the following skills :
A1 Analyze a system (real or virtual) or a problem (Level 1)
A3 Implement an experimental approach (Level 3)
A6 Report on an analysis or on a scientific approach (Level 3)
C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
C3 Evaluating, prototyping and simulating a system (Level 1)
C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 3)
C13 Considering technological and methodological innovation (Level 2)
B2 work, learn, progress autonomously (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :
- Mold design;
- Precedence constraints in machining;
- Isostatism;
- Cutting parameters.

To be able to :
- (A1, A3, A6, C2, C3, C6, C13, B2)
Design a mechanical manufacturing sequence by choosing a suitable material, designing a metal mold, respecting the geometric constraints, taking into account the economic constraints and machine capabilities

CONTENT

Mold design; precedence constraints in machining; Isostatism; cutting parameters

BIBLIOGRAPHY

Précis de construction mécanique. Projets-méthodes, Production, Normalisation ISBN : 2-09-194002-X
Fabrication par usinage. 2e édition ISBN : 978-2-10-051626-1
PRÉCIS DE FONDERIE MÉTHODOLOGIE, PRODUCTION ET NORMALISATION ISBN : 2-09-194019-4
Fonderie de précision à modèle perdu ISBN : 2-85330-100-1

PRE-REQUISITES

None